

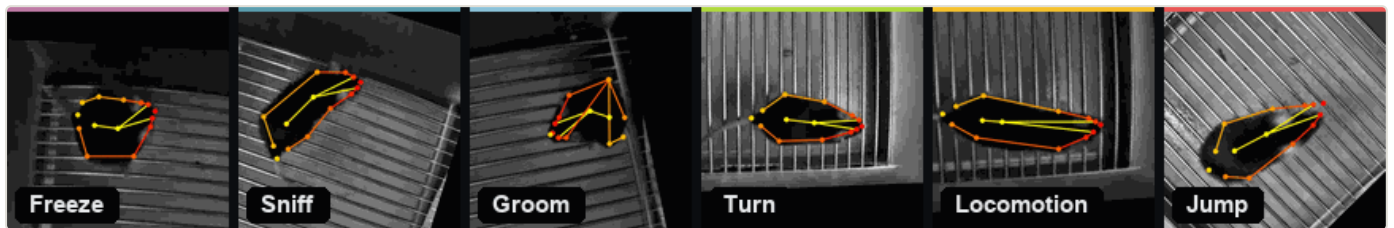
coping dynamics



EARLY-LIFE STRESS • BEHAVIOURAL SEQUENCING

Two coping profiles after the same early-life stress

Mice raised under limited bedding and nesting were fear-conditioned in adulthood, and their behaviour was sequenced with unsupervised keypoint-MoSeq rather than scored by an observer. The stressed group divides in two: a **vulnerable** subgroup whose sequences are more repetitive and less diverse than those of controls, and a **resilient** subgroup that does not differ from controls on any of those measures.



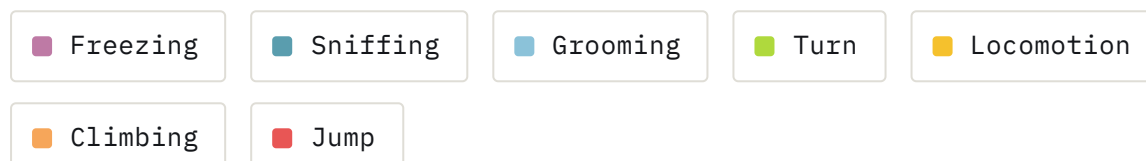
One representative keypoint-MoSeq syllable per behavioural cluster, egocentrically aligned, drawn from hash-verified archived clips.

THE REPERTOIRE

Sequencing behaviour instead of scoring it

The stress response is usually reported as a set of labels — fight, flight, freeze, fawn — assigned by an observer. The list has grown as observations accumulated, but the labels are categories imposed on a recording rather than quantities measured from it.

Keypoint-MoSeq segments pose tracking into short recurring motifs without supervision. Here the motifs map onto seven behavioural clusters, and each session becomes a sequence of those clusters at 250 ms resolution. The order of behaviour then becomes measurable, not only the amount of each behaviour.



Supplementary Video 1, grid view: all 25 cluster-mapped syllables playing at once, ordered and coloured by behavioural cluster. Each tile loops three representative pose-overlaid occurrences.

THE PARADIGM

Early-life stress, then fear conditioning in adulthood

Adversity in the first postnatal days, then a fear challenge at adulthood. The question is whether behaviour is organised differently when the threat returns, not only whether the stressed animals freeze more or less.

P2 – P9 Early-life stress Limited bedding and nesting, the chronic-adversity model of Rice et al., 2008.
P60 Fear conditioning A 30 s tone at 2.8 kHz and 80 dB, paired with a 2 s foot shock at 0.4 mA.
ANALYSIS Behavioural sequencing Pose tracking, then unsupervised keypoint-MoSeq syllables mapped onto seven behavioural clusters in 250 ms bins.

82	49	7	250 ms
animals clustered	litters	behavioural clusters	sequencing resolution

Every model in the paper is mixed-effects or GEE, grouped by animal, with litter composition and litter variance reported alongside (`statistics/litter_variance_components.csv`).

FINDINGS

What the sequences show

Four figures, in the order the argument is built. Each regenerates from the bundled data with `scripts/run_all.py`, the plotted values are exported next to it, and each links to the vector PDF at full resolution.

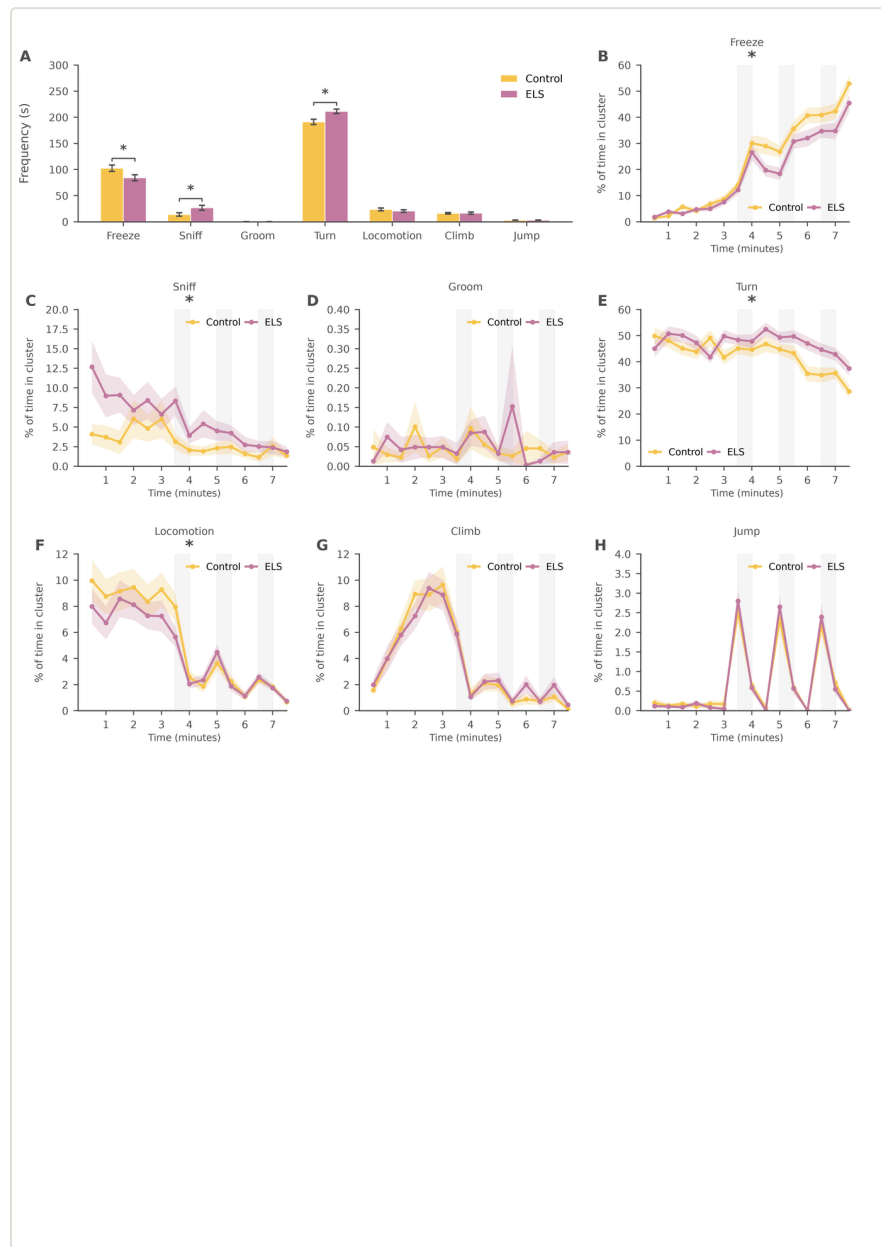


FIGURE 3

Behaviour frequencies and time courses differ between groups

Group differences appear both in how often each behaviour occurs and in how those frequencies change across the session, rather than as a constant offset between groups.

[figure_source_data/figure3.csv](#) · [Full-resolution PDF](#)

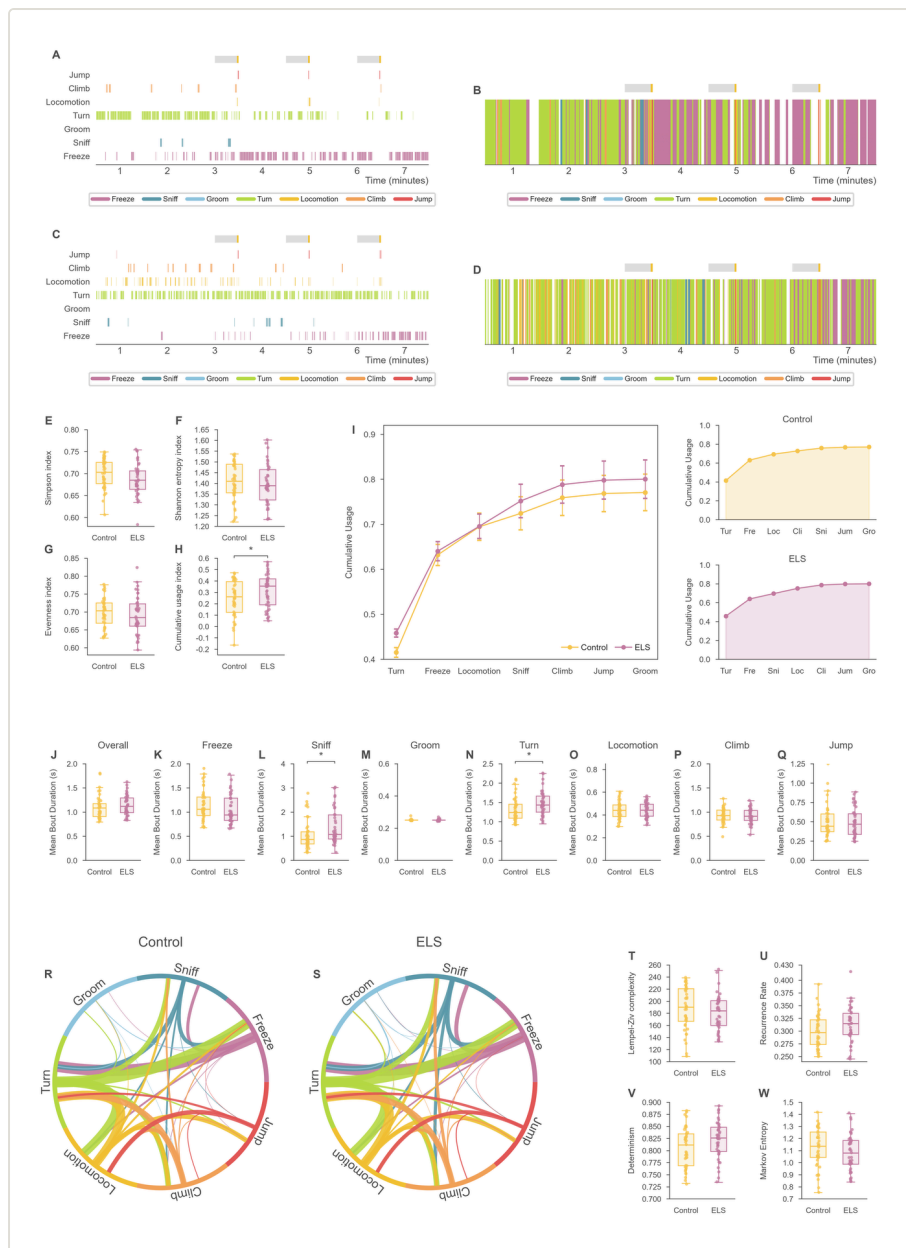


FIGURE 4

Stressed animals hold active behaviours longer

Sniffing bouts are 0.32 s longer and turns 0.16 s longer than in controls ($p = 0.029$ and 0.020). Recurrence and determinism are higher in the stressed group but do not reach significance ($p = 0.058$ and 0.075).

statistics/stats_figure4_bouts_MixedLM.csv · Full-resolution PDF

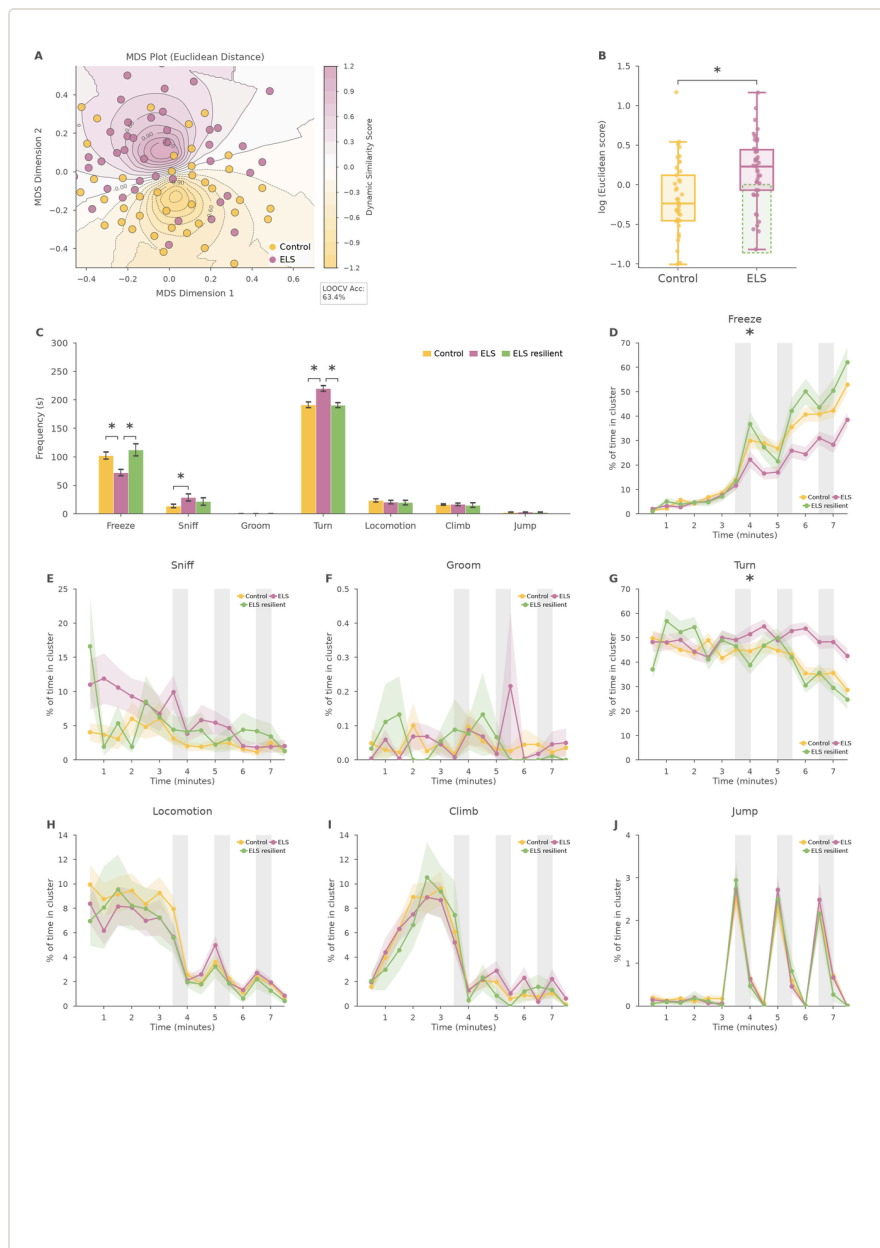


FIGURE 5

A dynamics score splits the stressed group in two

Each animal is scored by how close its behavioural dynamics sit to the control distribution. Clustering on that score separates **29 vulnerable** from **12 resilient** stressed animals, without using the outcome measures.

[statistics/animal_dendrogram_clusters.csv](#) · [Full-resolution PDF](#)

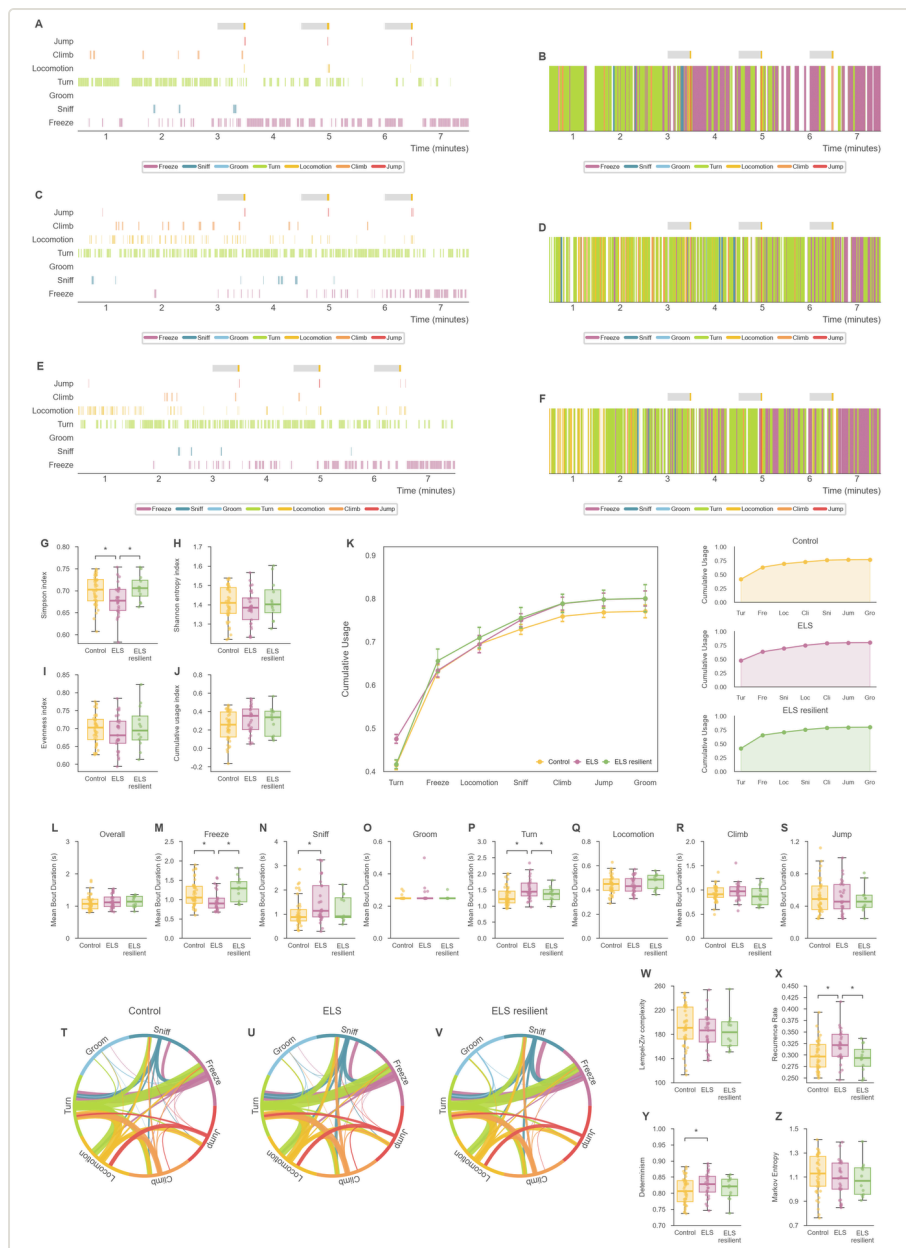


FIGURE 6

The split sharpens the group differences

Vulnerable animals return to the same states more often and use a narrower repertoire than controls. Resilient animals differ from the vulnerable group and not from controls.

[statistics/fig6_*_resilience_stats.csv](#) · [Full-resolution PDF](#)

TWO PROFILES

Vulnerable and resilient animals against controls

Pooled across all stressed animals, the sequence measures move in the expected direction but stop at trends. Split by coping profile, the picture is consistent: **on every measure that separates vulnerable animals from controls after FDR correction, resilient animals do not differ from controls.**

SEQUENCE MEASURE	ELS VULNERABLE VS CONTROL	ELS RESILIENT VS CONTROL
Recurrence rate	+0.026 q = 0.005 differs	-0.008 q = 0.69 n.s.
Simpson diversity	-0.022 q = 0.013 differs	+0.007 q = 0.77 n.s.
Freezing bout duration	-0.18 s q = 0.028 differs	+0.08 s q = 0.73 n.s.
Sniffing bout duration	+0.42 s q = 0.042 differs	+0.09 s q = 0.73 n.s.
Turn bout duration	+0.22 s q = 0.028 differs	+0.02 s q = 0.84 n.s.

Mixed-effects coefficients grouped by animal, with Benjamini–Hochberg FDR correction within each metric family; bout coefficients are in seconds. The resilient-versus-vulnerable contrast is significant in the same direction for recurrence (q = 0.005) and Simpson diversity (q = 0.010). Every value here is read from the shipped tables `statistics/fig6_transition_resilience_stats.csv`, `fig6_diversity_resilience_stats.csv` and `fig6_bout_resilience_stats.csv`, which the `rebuild` regenerates.

REPRODUCE

Rebuilding the analysis

Every figure, table, statistic and report on this page rebuilds from the data in the repository. There are no hidden inputs and no download step: a clone contains everything needed.

```
# install the locked environment
uv sync --locked

# verify the shipped artifacts against MANIFEST.csv
uv run python scripts/check_reproducibility.py

# rebuild everything – the tree ends clean
uv run python scripts/run_all.py
```

Hash-checked artifacts

MANIFEST.csv carries a SHA-256 for every artifact. A rebuild that changes one byte fails the check, on Windows and on Linux alike.

Statistics fitted once

Each model is fitted in one place and exported; scripts/check_manuscript.py pins every statistic quoted in the manuscript to the shipped report.

Companion classifier

tools/behavior_dlc_classifier/ ships the trained DeepLabCut networks, demo videos, tests and a guide, so new recordings can be tracked and classified without training anything.

Coping Dynamics Sequencing — Jeniffer Sanguino-Gómez and colleagues, Brain Plasticity group, SILS-CNS, University of Amsterdam.

Code is MIT-licensed and the bundled data CC BY 4.0. Citation metadata lives in CITATION.cff — use GitHub’s “Cite this repository”. An archived, DOI-minted release accompanies the manuscript at submission.

github.com/antonio-lozano/coping-dynamics-sequencing